which gives in general a nonlinear system of N_c equations in N_c unknowns to be solved for the position vector $\boldsymbol{q}^*$ associated to mechanical equilibrium.

At the equilibrium point, we can linearize the system of differential equations by computing the Hessian matrix, then we find

$$\nabla U(\boldsymbol{q}) \approx U''(\boldsymbol{q}^*)(\boldsymbol{q} - \boldsymbol{q}^*).$$

Then, letting $\delta\boldsymbol{q} = \boldsymbol{q} - \boldsymbol{q}^*$, $\delta\boldsymbol{p}$ represent small deviations from the equilibrium point at $(\boldsymbol{q}, \boldsymbol{p}) = (\boldsymbol{q}^*, \boldsymbol{0})$, we have

$$\frac{\mathrm{d}\delta\boldsymbol{q}}{\mathrm{d}t} = \boldsymbol{M}^{-1}\delta\boldsymbol{p},$$
$$\frac{\mathrm{d}\delta\boldsymbol{p}}{\mathrm{d}t} = -U''(\boldsymbol{q}^*)\delta\boldsymbol{q}.$$

At the minimum of the potential energy, U'' is a positive definite symmetric matrix. The eigenvalues of the matrix

$$\boldsymbol{A} := \begin{bmatrix} \boldsymbol{0} & \boldsymbol{M}^{-1} \\ -U''(\boldsymbol{q}^*) & \boldsymbol{0} \end{bmatrix}$$

are therefore all purely imaginary ($\pm i\Omega$, $\Omega \in \mathbb{R}$). Associated to each eigenvalue pair we have a pair of complex conjugate eigenvectors $\boldsymbol{\xi}, \bar{\boldsymbol{\xi}}$ and also a pair of solutions which can be written in the complex form

$$z(t) = ae^{i\Omega t}\boldsymbol{\xi} + be^{-i\Omega t}\bar{\boldsymbol{\xi}},$$

(a, b complex coefficients), or recast in real form as (α, β real coefficients):

$$z(t) = \alpha[\sin(\Omega t)\mathrm{Re}(\boldsymbol{\xi}) + \cos(\Omega t)\mathrm{Im}(\boldsymbol{\xi})] + \beta[\cos(\Omega t)\mathrm{Re}(\boldsymbol{\xi}) - \sin(\Omega t)\mathrm{Im}(\boldsymbol{\xi})].$$

These are referred to as the "normal modes" of oscillation of the system (Fig. 1.20 illustrates the normal modes for a linear molecule).

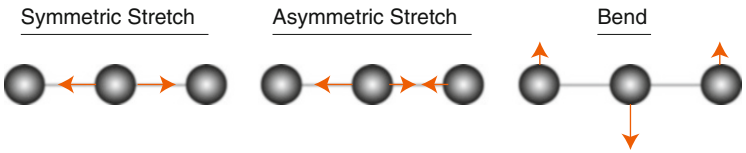


Fig. 1.20 Three normal modes of a linear molecule (i.e., a molecule in which the atoms, at minimum energy, are arranged on a line)